Manchu Learning: A Manchu-English Machine Translation Project

DSAN6600 Neural Networks and Deep Learning

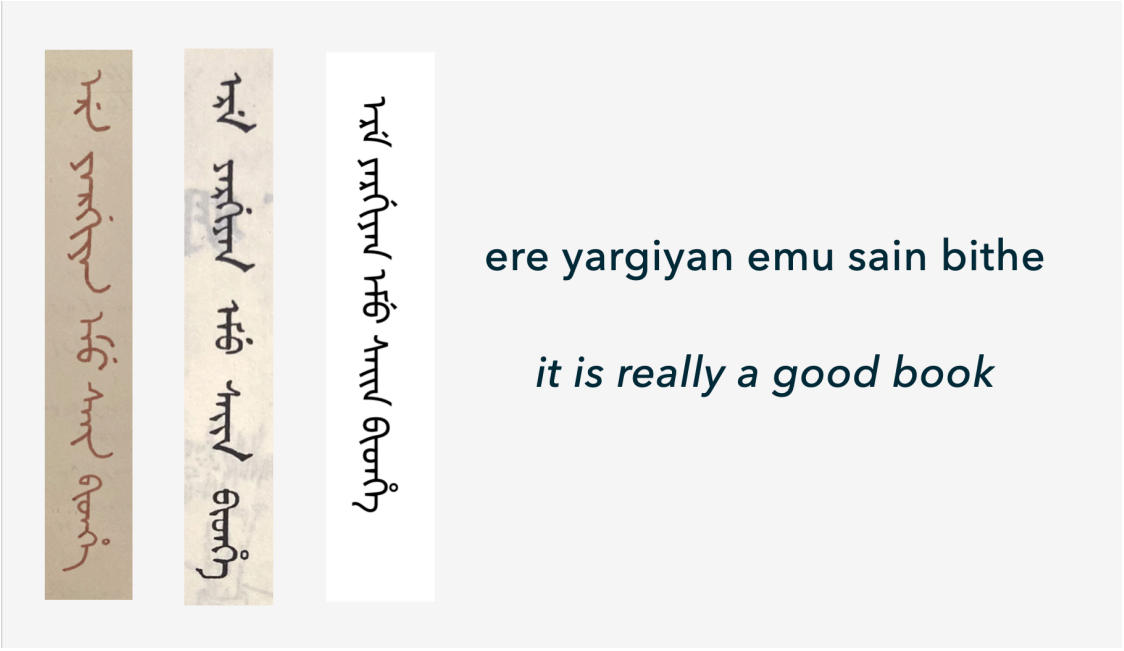
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Introduction

This project is an attempt to build a Neural Machine Translation model for the challenging task of Manchu-English translation.

- Keywords: Neural Machine Translation, Data Curation, In-Context Machine Translation, Endangered Language

The Language Manchu



Our Goal and the Challenges Along the Way

1. Language in Danger: Manchu is often treated like Latin, mainly learned to read historical documents. Unlike Latin, however, it still has a spoken tradition today. In the **Qapqal Xibe Autonomous County** in Xinjiang, where its use is part of deliberate preservation efforts, taken by the Xibe people in the name of the Xibe (Sibe) language
2. Digitization challenges: As some may have observed, the Manchu writing system is a distinct one, comparing to some common writing systems in the world. First, the direction, in **columns** and from left to right, similar to Arabic rotated 90 degrees counter-clockwise. Second, Manchu script (alphabets) can be described as **IMFI** script which letters can take on more than four different shapes depending on where they appear in the word or whether they are in isolation. This complicates the picture for digitization work using Optical character recognition (OCR), due to the extra work on letter segmentation.
3. Transliteration
4. Translation

Literature Review

In this section, I want to briefly talk about the three papers that introduced me to the task of Manchu machine translation and helped shape the methods and approaches in this project. These studies, in their distinctive ways, engage the Manchu Language revitalization effort within the broader dialogue of Neural Machine Translation in low-resource settings.

[Seo et al., *"Mergen: The First Manchu-Korean Machine Translation Model Trained on Augmented Data."*]Seo et al., *"Mergen: The First Manchu-Korean Machine Translation Model Trained on Augmented Data."*[1]

This paper introduces us to the challenges of developing the typically data-driven neural machine translation models for Manchu, which suffers from (an extreme) scarcity of parallel data, not only in the Manchu-Korean case, but more generally. The authors thus come up with an innovative data augmentation strategy based on **word replacement**, where lexical similarity is determined using GloVe (Global Vectors for Word Representation) embeddings of dictionary items. By augmenting the parallel corpus in this way, they witness a significant improvement in the BLEU (Bilingual Evaluation Understudy) score of their seq2seq (Sequence-to-Sequence) model, from 0 to 38, as the number of sentence pairs grew from 20K to 179K.

[1] Seo et al. (2024)

[Lee et al., *"ManNER & ManPOS: Pioneering NLP for Endangered Manchu Language."*]Lee et al., *"ManNER & ManPOS: Pioneering NLP for Endangered Manchu Language."*[2]

This study, from the same Seoul National University (SNU) team, continues their line of work and with their data augmentation technique, they further contribute by training the Manchu Langauge Model **manchuBERT** from scratch and releasing the model on Hugging Face. The manchuBERT langauge model serves as the starting point for developing NER (name entity recognition) and POS (part-of-speech) tagging for Manchu. The strong performance of these downstream NLP models, in turn, provides evidence that manchuBERT langauge model has indeed learned to be a robust encoder for Manchu.

[Pei et al., *Understanding In-Context Machine Translation for Low-Resource Languages: A Case Study on Manchu*]Pei et al., *Understanding In-Context Machine Translation for Low-Resource Languages: A Case Study on Manchu*[3]

This third paper also discusses the prospects of low-resource machine translation, but approaches the problem from a different angle. Rather than focusing on model architectural modifications or data augmentation techniques, the authors consider how **human-curated linguistic knowledge**, such as dictionaries and grammar books can contribute in this picture. Specifically, they explore how linguistic resources can be instilled into pre-trained LLMs as such models can leverage their in-context learning (**ICL**) capability for the translation of languages they were not trained on. Through a series of designed experiments, they systematically examine the effects of different types of linguistic information. The authors arrived at the conclusion that dictionaries that comprehensively cover lexical meanings, suffix explanations, and morpheme collocation information can effectively improve translation quality, as can parallel examples.

Data

Data Collection

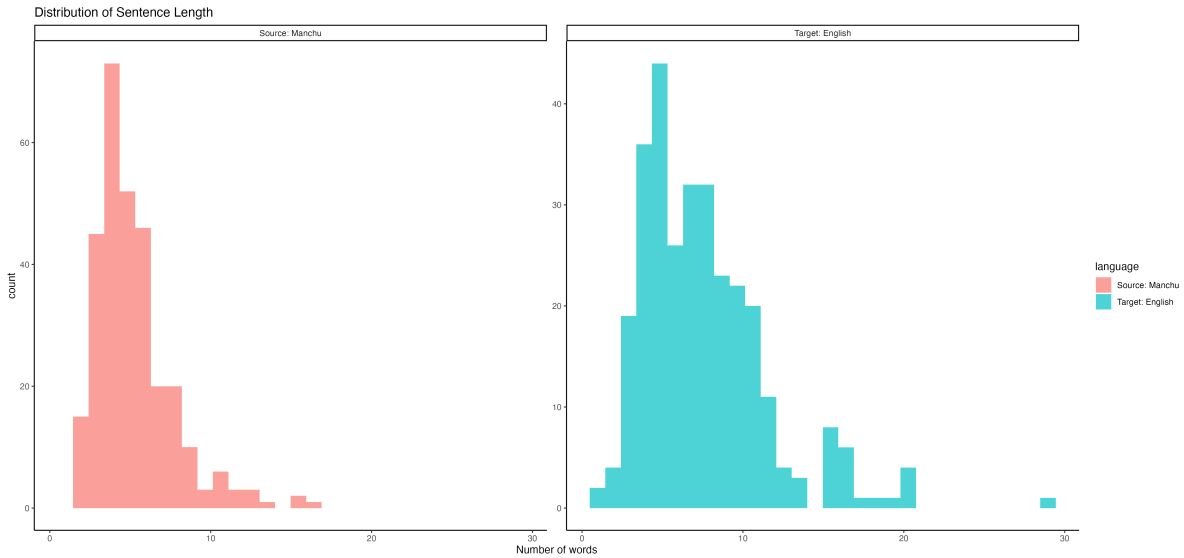
The data used in this project is a small but carefully examined Manchu-English parallel dataset. I chose the renowned novel ***The Dream of the Red Chamber*** written in classical Chinese by Cao Xueqin (published in 1790s Qing China)[4], in specific, the novel's Manchu translation((1993)) and English translation (Cao (2006)) I am hoping that translators of this canonical work would place stronger emphasis on fidelity to the original text, thus, expecting it to be relatively easier for me to find sentence-level alignments that would support my endeavor to build Manchu-English sentence pairs

Here, I present some key distributions of my dataset:

[2]Lee et al. (2024)
[3]Pei, Liu, Lin, Yvon, and Schuetze (2025)
[4]the novel was written in "Written Vernacular Chinese", but still not that as explicit as modern Chinese

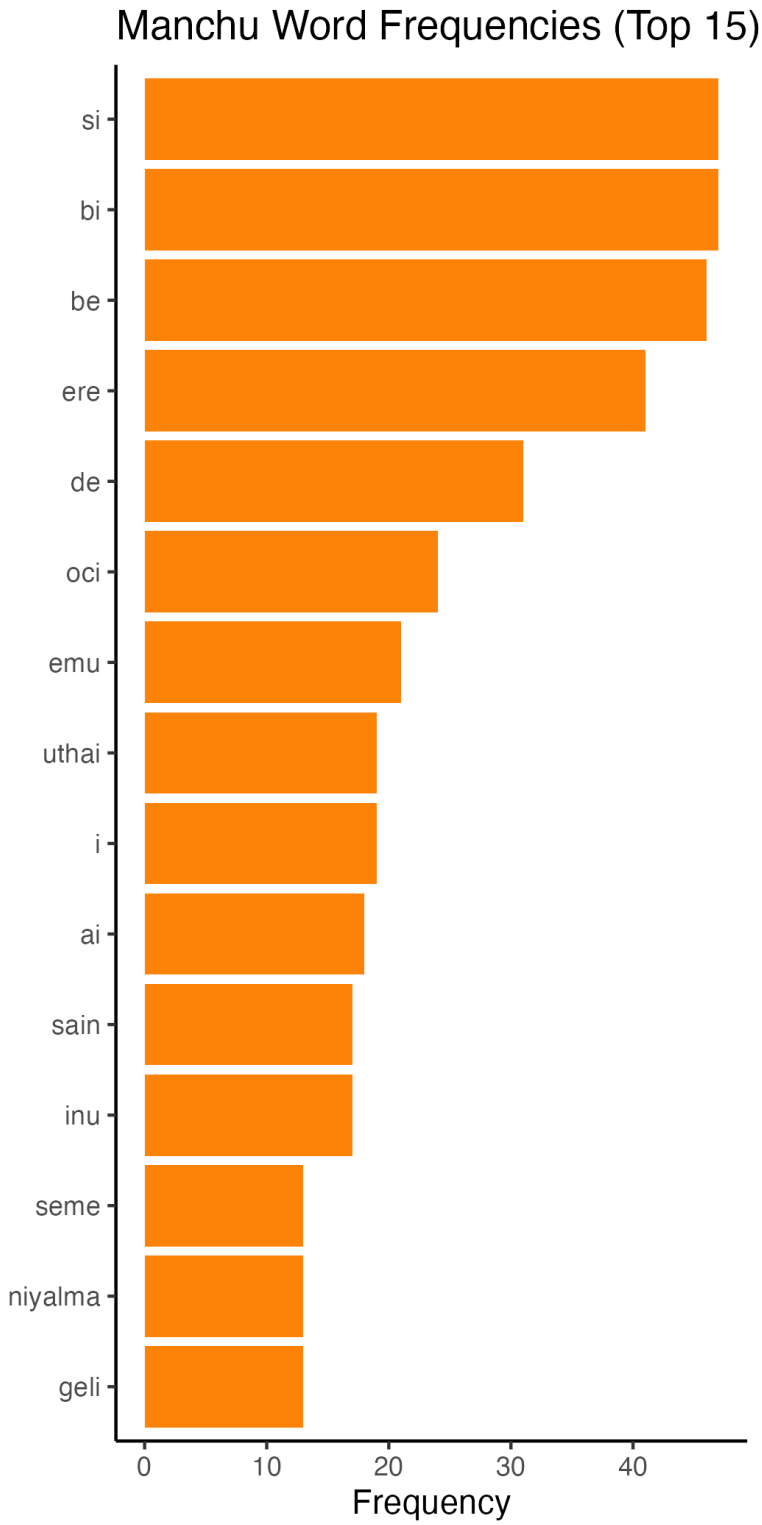
Sentence Length:

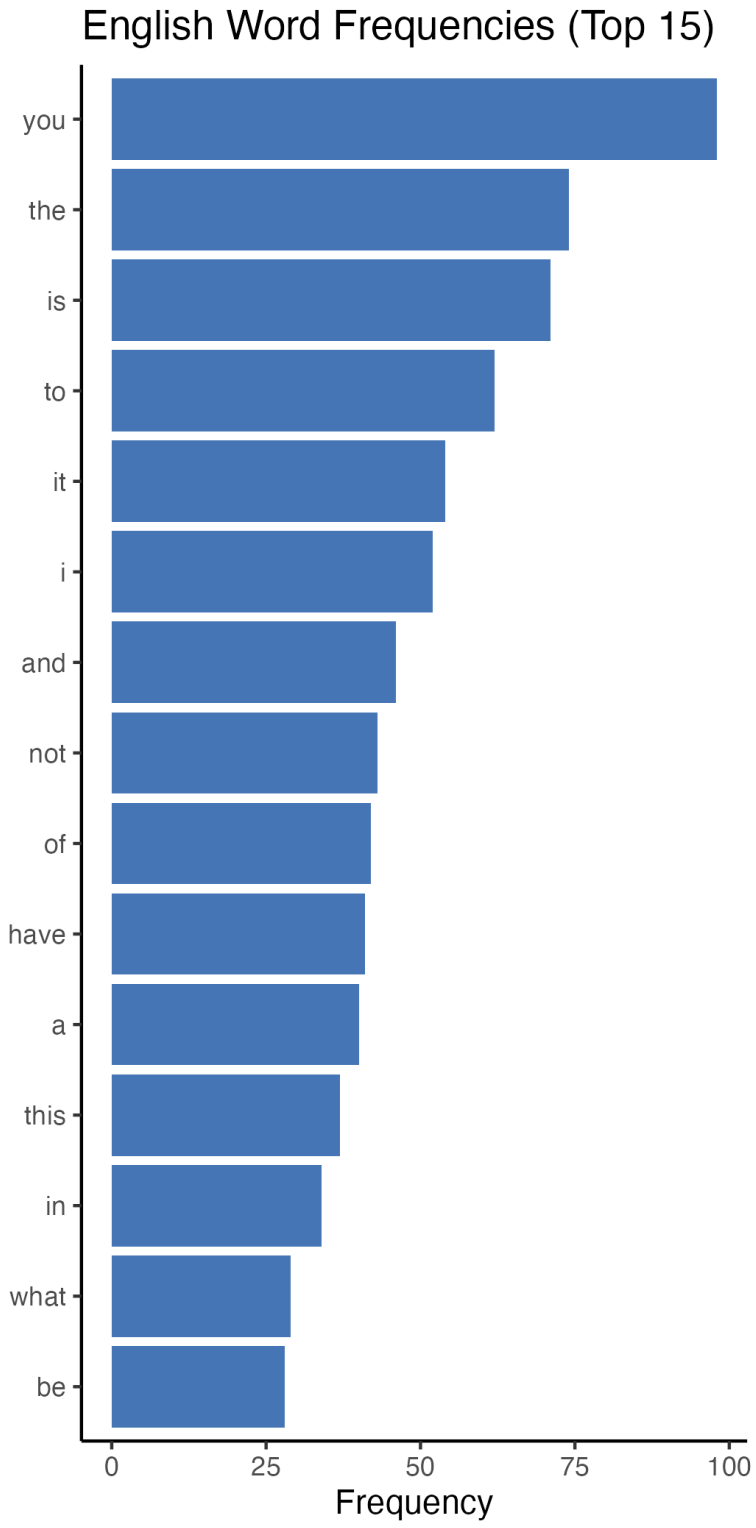
- Manchu: 5.38 words per sentence on average
- English: 7.73 words per sentence on average



Vocabulary Size:

- Manchu: 700 distinct words
- English: 669 distinct words


Manchu Word Frequencies (Top 15)
si
bi
be
ere
de
oci
emu
uthai
i
ai
sain
inu
seme
niyalma
geli
0
10
20
30
40
Frequency


English Word Frequencies (Top 15)
you
the
is
to
it
i
and
not
of
have
a
this
in
what
be
0
25
50
75
100
Frequency

Data Split

I reserved 12.5% for validation (37 rows), and 38 rows for testing, leaving 225 rows for training. Given the very limited size of our data, and the often large number of training parameters in our neural models, regularization and monitoring validation loss are both very crucial concerns in the training process.[5]

Data Preprocessing

Text cleaning and normalization

For simplicity, I removed all punctuation, and all sentences were converted to lowercase.

> **i** The issue of Transliteration Alignment
>
> I faced challenges in the data preparation phase is around the issue of transliteration, and I found it worth recording, so I would like to share it here. The transliteration system that I was using initially for my self-curated dataset was the **Möllendorff-Norman** system, because it is the one our Manchu professor taught, and is also one of the most used systems.
> Later, in an attempt of harvesting from the pre-trained manchuBERT, I found in the *Mergen (2024)* paper statements that they use **Abkai transliteration**. I converted my dataset into Abkai, only to find out that the authors do not fully adopt this transliteration system. To clarify this, and because the datasets disclosed in their github repository are the test sets, I ran a small experiment using manchuBERT as a masked language model. After inputting some sentences with mask tokens, the outputs of manchuBERT suggested that they actually use a hybrid system for transliteration. For example, the model represents the sixth Manchu vowel using `v` rather than the Norman `ū`; uses `x` instead of the Norman `š`, which seems to be aligning to the Abkai system. At the same time, it employs Norman `c` rather than the Abkai-preferred `q`.
> Based on these observations, I re-romanized the Manchu side of my data to align with what manchuBERT learnt. I think this transliteration alignment process that I underwent actually highlights a non-trivial challenge in obtaining and integrating online Manchu resources: when without a standardized transliteration system, languages that use non-Latin scripts are going to suffer, or will require additional attention in the data pre-processing stage.

Tokenization

[5] a better and definitely doable robustness test is to implement k-fold cross validation so as to eliminate the effect of data splitting for small sample

Different ways for tokenizations are implemented across my four MT attempts. I would like to provide a brief illustration in this section before we go into the models.

In *Attempt 1*, tokenization was done by splitting our sentences by whitespace, resulting in word-level tokenization.[6] In *Attempt 2*, to correctly map to manchuBERT, I use the AutoTokenizer from Hugging Face Transformer library to retrieve the tokenizer of our pre-trained model. And as stated in the paper, the tokenization they applied is **WordPiece**; in the English output part the bert-base-ubcased tokenizer was applied (Devlin et al. (2018)).[7]In *Attempt 3*, I continued to use the manchuBERT tokenizer for Manchu source sentences; the English counterpart used the SentencePiece tokenization, which completes the **subword** segmentation, and I had to additionally specify the tokenizer to translate into English. Finally, in *Attempt 4*, I carried out a similar practice as of the Pei et al. (Pei, Liu, Lin, Yvon, and Schuetze (2025)), I wrapped up the sentences with corresponding prompts and fed directly to google colab AI.

Approaches

Attempt 1: 2 GRUs – Seq2seq Network with Attention

This is the baseline model I used, and I directly referred to the Pytorch tutorial for the implementation from scratch("NLP From Scratch: Translation with a Sequence to Sequence Network and Attention — PyTorch Tutorials 2.9.0+cu128 Documentation" (n.d.)), and also because it has a similar structure to the descriptions in the Mergen paper(Seo et al. (2024))

Model Architecture:

- The sequence-to-sequence model involves two RNNs, carried out as **Gated Recurrent Units**. Our first GRU did the encoder job, and the second GRU has a twist of the attention weights calculation in the decoding.[8]

Hyperparameters:

- Learning Rate = 0.001
- Embedding size = 256
- Hidden Size = 512
- Dropout Rate = 0.2
- Epochs = 20

[6]with four additional tokens, , , ,
[7]with manually handling and
[8]the attention mechanism is the **additive Bahdanau attention**, one way of looking at this mechanism, in the context of neural machine translation, is that it helps in many ways to the ability to deal with word ordering of our languages, and order the varying lengths of the sentences.

Attempt 2: Frozen ManchuBERT + LSTM Decoder

Model Designs:

- My second attempt reflects the representation-level transfer learning, which I used the `Automodel` class from transformers library to load the **manchuBERT** model configuration from Hugging Face, and then I used the method `from_pretrained()` to acquire the model parameters. As provided in the documentation (Seo et al. (2024)), manchuBERT is a language model trained on more than 5.2 M monolingual Manchu sentences[9] The embedding vector is of 768 length. With the help of manchuBERT our raw Manchu input can be transformed into rich and context-aware representation space, as an encoder I pair it with an LSTM decoder for our downstream translation task.

Hyperparameters:

- Learning Rate = 2e-4
- Embedding size = 256
- Hidden Size = 512
- Number of Layers = 2
- Dropout Rate = 0.2
- Epochs = 50

Attempt 3: manchuBERT & mBART

Model Designs:

- In this third attempt, I introduced new dimensions to the model. The encodings are passed to the **mBART**,[10] the pre-trained, **end-to-end** multilingual to English model there to support the next step that takes encoded Manchu sentences to sensical English.(Liu et al. (2020))[11]

Hyperparameters:

- Learning Rate = 3e-5
- Epochs = 30

[9] 195,611 monolingual Manchu sentences before data augmentation

[10] One of the key contributions of the mBART model is that it is proven to be conducive to translation of even the languages unseen by the model to fluent English sentences

[11] https://huggingface.co/transformers/v4.4.2/_modules/transformers/models/mbart/tokenization_mbart50_fast.html

Attempt 4: Norman Dictionary + Google Colab AI – In-Context Machine Translation

Experiment Designs

- Our last attempt was primarily inspired by the ICL Machine Translation paper[12] I explored the linguistic resource *The Comprehensive Manchu-English Dictionary* from Jerry Norman(Norman (2013)) Step by step, I first parse the dictionary into `item-meaning` pairs of 21K, and I retrieve the vocabularies from our source Manchu sentence and incorporate the dictionary information as part of the translation prompt
- One more special account for Attempt 4 is that rather than directly enciphering our source to eliminate the prior knowledge of our LLM, I introduced some controls to the experiment by speicifying that the task is Manchu-to-English or not specifying the target language, or by creating a substitute such as "Anchum" language in the prompt.

Prompt example:

```
Please help me translate the following input sentence into English
For the translation task, you are given the word by word mapping
from the {source language} words to the {target language} words

        Please try your best to translate, it's okay if
        your translation is bad. Do not refuse to try it.
        I won't blame you.

IMPORTANT:
- Output the input sentence's English translation ONLY.

MINI: genitive of bi: my, of me
OKTO: drug, medicine; gunpowder; dye; poison
SINDE: dative/locative of si
AI: what? which?; hey!
DALJI: relation, bearing, connection

### TRANSLATION ###
Input: mini okto omire omirakvngge sinde ai dalji
English:
### END ###
```

```
Source: mini okto omire omirakvngge sinde ai dalji
What connection does my medicine to be consumed have to you?
```

[12] Pei, Liu, Lin, Yvon, and Schütze (2025)

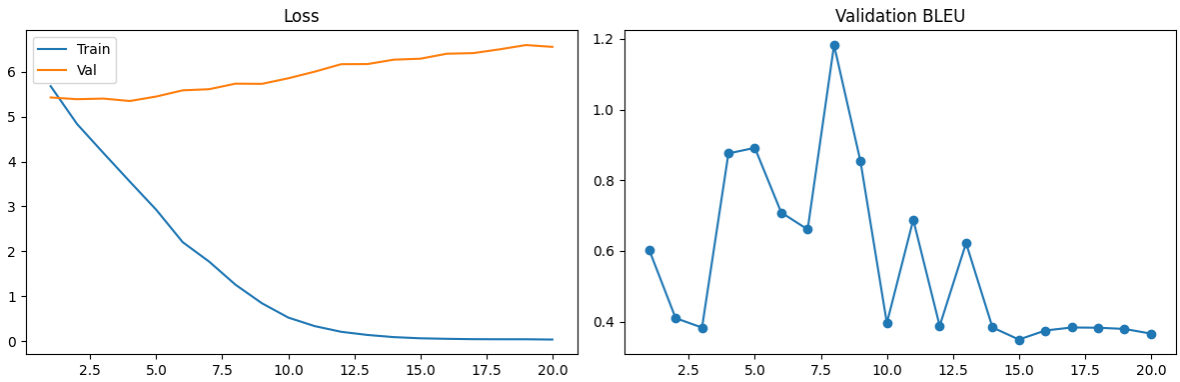
Results

Evaluation Metrics

1. Automatic Evaluation: Bilingual Evaluation Understudy **BLEU** : The evaluation metric BLEU is implemented using the Python package `sacreBLEU` which, we can think of as **modified n-grams precision** plus a **brevity penalty**, as it is intuitive, for that a translation task does not have a single ground truth or absolute correctness. Thus, in the evaluation of machine translation models, test cBLEU is the most used metric.
2. Sentence Level BLEU: from the sacrebleu API, with the argument `effective_order = True` we can acquire a variant of the original corpus-level BLEU
3. Validation loss and BLEU: Both validation loss and validation BLEU are used to determine multiple model checkpoints[13]

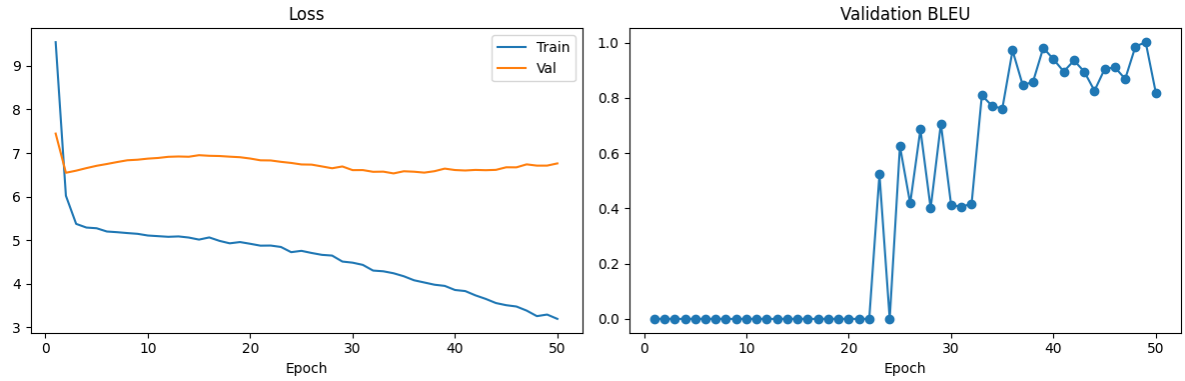
Training Reports

Plot 1: Attempt 1 Training Trace

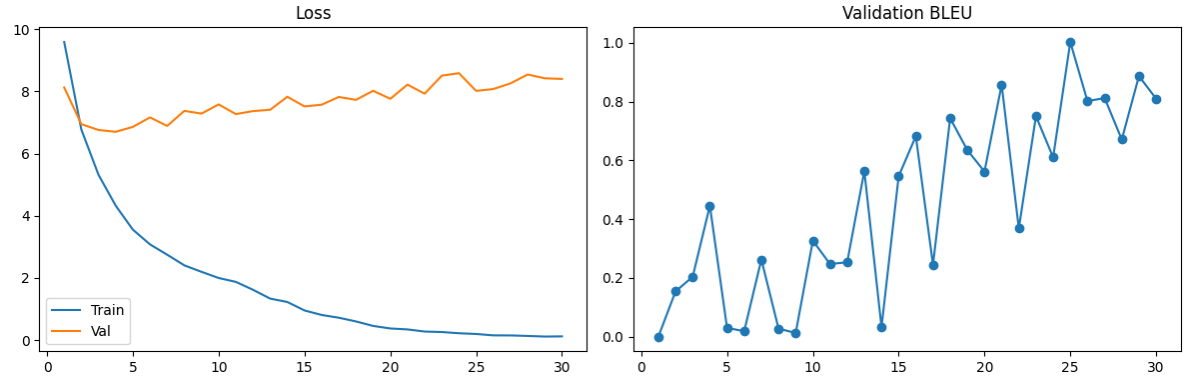


Plot 2: Attempt 2 Training Trace

[13]In regard to the small size of our data, validation loss alone can interrupt training signals and can often prevent the model from optimize, thus, to strike a balance between underfit and overfit, I save three models for each attempt and also download there predictions on the test set for later human evaluation of translation quality



Plot 3: Attempt 3 Training Trace



Across all three NMT models, the training loss decreased monotonically, indicating these algorithms' ability to fit our training data. In contrast, as for the validation loss curves, three models all achieved the lowest validation loss before epoch 5, which reflects the contraints imposed by the size of our corpus. That same data size issue can explain part of the unstable and spiky behavior for validation BLEU scores over the training. Notably, in our attempt2, the manchuBERT-LSTM arrangement, there seems to be an increasing trend in the validation BLEU score.

Evaluation Metrics

Model / Approach	Test BLEU (Best Val BLEU)	Test BLEU (Best Val Loss)	Test BLEU (Longer Training)
Seq2Seq (2 GRUs)	2.08	0.72	0.94
manchuBERT + LSTM	1.38	1.27	1.44
manchuBERT + mBART	1.86	0.47	1.86

To understand these BLEU scores,[14] we can first look at the results scores from the first three attempts, which involve model training. Under the conventional validation set approach for model selection, choosing the best model based on validation loss, the manchuBERT + LSTM model performs best. If we allow another way of assessing the model's ability to extrapolate, the BLEU scores evaluated on the models with the best BLEU score, then it is the baseline seq2seq network that outperforms other two. This aligns with our understanding of the relationship between data size and the size of training parameters.

I observed that in theory, the combined manchuBERT + mBART model arrangement made sense. However, in implementation, I wrestled with the tokenization alignment quite a bit only to get it working and the training was very slow. This to a certain extent points me to an understanding of model flexibility, that the model architecture of my third attempt was a bit too grand and sophisticated with respect the 300 language pairs that I had at hand.

Moving towards my fourth attempt, I was interested in examining, provided linguistic knowlege and lexical meanings of the words in a sentence alone, LLM's capability to do translation on languages that they were not explicitly trained on. Thus, I design an experiment with hope that I can to some extent eliminate the prior knowledge of Manchu of LLM. Specifically, I referred to the Pei et al. paper(Pei, Liu, Lin, Yvon, and Schütze (2025)), and performed a comparison between four conditions created by different prompt configurations:

1. Directly Translation with no dictionary resource and no source language information
2. The source language is not specified but dictionary entries are provided
3. Provide deliberately misnamed source language `Anchum` and the right dictionary entries
4. Source language is Manchu directly specified with dictionary entries

****	Explicitly Manchu-English + dict	Misnaming source as Anchum + dict	Source language not specified + dict	Direct translation without dictionary
In-context LLM + Dictionary	4.47	1.65	1.77	1.44

We can a spot a distinctively higher test BLEU score for the condition which we not only provide lexical mappings in the prompt but also specify explicitly that the task is to do Manchu-English translation. While the performance drops when there is misinformation or no information about the source language. However, from the generated translations, we can see that LLMs are good at forming fluent English sentences given a bag of words and meanings. It can be suggested that the BLEU score is underestimated due to the abundant

[14]Test BLEU scores reported here correspond to the model perfomance on the held-put test set at check points selected by highest validation BLEU, by lowest validation loss, and at checkpoint for longer training (epoch 15)

verb conjugations or suffixations in Manchu grammar, one concrete example is that when a word in a Manchu sentence does not have a dictionary entry, the absence of a English definition could easily causes the translation quality to drop.

Translation Sample

Although BLEU is designed to be a more robust way to evaluate machine translation, comparing to using the cross entropy loss alone, it's still operating on the n-grams mindset. Regarding this limitation, I go for the most intuitive way to examine translation quality, which is the translation itself for the test set by different models. This leads me to some interesting observations which would serve a good ending for the evaluation section.

We can see that Attempt 4 (in-context LLM + Dict), generally yields decent translations, nevetheless this approach is vulnerable when words in a sentence did not find a mapping in dictionary, such as in the case of ID 4, "loo tai tai"—a Manchu transliteration of a Chinese word meaning old lady, does not appear in the dictionary, but they are mismatched to 'LOO: prison; gong; cf. lo; see lomi', 'TAI: platform, terrace, stage', which drives the translation off the correct path. Interestingly, for the same ID 4 sentence translation, Attempt 3(manchuBERT + mBART) somewhat captured the meaning of loo tai tai in the sentence. The cases that the translation of Attempt 3 are underestimated, ID 16 is one example and it almost captures the first half of the sentence, but later falls into decoder degeneration.

Attempt 1(2 GRUs) often produces fluent and syntactically well English translation, and the translation often involve sensical phrases such as "at all" in ID 4 and "at this early hour" for ID 20 translation, which aligns with part of the sentence pretty well.

```
==============================
ID: 4
==============================
SOURCE (Manchu):
jiduji loo tai tai fulu kai

REFERENCE (English):
your venerable ladyship
---- attempt1_seq2seq_gru ----
Sentence BLEU: 0
if you have been at all
---- attempt2_manchubert_lstm_frozen ----
Sentence BLEU: 0
we will be to the in of the
---- attempt3_manchubert_mbart ----
Sentence BLEU: 0
```

```
where it is it will dear senior me
---- attempt4_incontext_dict ----
Sentence BLEU: 0
after all the prison platforms are truly excellent

==============================
ID: 16
==============================
SOURCE (Manchu):
si absi erei gese sain arahabi

REFERENCE (English):
how is it you have written them so well
---- attempt1_seq2seq_gru ----
Sentence BLEU: 7.45
you have been
---- attempt2_manchubert_lstm_frozen ----
Sentence BLEU: 17.87
how is it it it
---- attempt3_manchubert_mbart ----
Sentence BLEU: 19.49
how is it that you how is it
---- attempt4_incontext_dict ----
Sentence BLEU: 12.26
how well you have made it like this
```

```
==============================
ID: 20
==============================
SOURCE (Manchu):
enteke erde uthai feksime jifi ainambi

REFERENCE (English):
what have you run over to do at this early hour
---- attempt1_seq2seq_gru ----
Sentence BLEU: 26.3
if you have been at this early hour
---- attempt2_manchubert_lstm_frozen ----
Sentence BLEU: 4.41
how is it that you have it
---- attempt3_manchubert_mbart ----
Sentence BLEU: 7.5
```

```
you have you but you have come back again for me
---- attempt4_incontext_dict ----
Sentence BLEU: 4.52
why are you coming leaping so early in the morning
```

Discussion

In general, all machine translation attempts in this project show low BLEU scores, which is expected in terms of the obvious discrepancy between the millions of parameters these models have and the very modest size of my dataset. **Low-resource NMT** has become a very popular reseach area recently, this project explored many of the proposed remedies, leaving others not reached, in doing so to some degree tested the practical boundaries of what counts as low-resource language. Many languages described as "low-resource" in the NLP literature still possess orders of magnitude more data than Manchu, which has almost no modern digital presence.[15]

However, this practice and seeing it as a concrete appreciation of the obstacles of data scarcity in a NMT model, is meaningful to me because I finally get to go from not just vaguely think about how I can use data science to preserve Manchu but to actually get a neural machine translation model working. And throughout this project, I was able to familiarize myself with where the exisiting resource are, and started thinking about what resource I can potentially introduce. In addition, getting to meet these communities dedicating to Manchu preservation with their research effort was in itself very instructive and motivating.

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[15] low relatively to French-English, Spanish-English pairs

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